1 Data Generation

2 Treatments: Tmt1: 50, Tmt2: 55, or both are 50 (will do 1000 sets when equal, and 1000 when not equal)

10 Subjects per Treatments

5 Reps per Subject

100 data points

AR1 Covariance: $\sigma_e^2 \begin{pmatrix} 1 & \rho & \rho^2 & \rho^3... \\ \rho & 1 & \rho & \rho^2... \\ \rho^2 & \rho & 1 & \rho... \end{pmatrix}$

4 covariance structures (the following are the diagonal components):

- Indepdenent: $\sigma_e^2 I$
- CompSymm: $\sigma_e^2 = 10, \sigma_s^2 = 12$
- AR1: $\sigma_e^2 = 10, \rho = .8 : \begin{pmatrix} 10 & 8 & 6.4 & 5.12 & 4.1 \\ 8 & 10 & 8 & 6.4 & 5.12 \\ 6.4 & 8 & 10 & 8 & 6.4 \\ 5.12 & 6.4 & 8 & 10 & 8 \\ 4.1 & 5.12 & 6.4 & 8 & 10 \end{pmatrix}$
- Unstructured: $\begin{pmatrix} 15 & 10 & 12 & 12 & 2 \\ 10 & 18 & 5 & 13 & 8 \\ 12 & 5 & 20 & 5 & 3 \\ 12 & 13 & 5 & 15 & 10 \\ 2 & 8 & 3 & 10 & 12 \end{pmatrix}$

Test using AIC/BIC (groups will pick one). Pick a covariance structure.

Generate data using each of the covariance structures. Then test each dataset using all 4 covariances. Then use the AIC/BIC to pick one.

Repeat this process 1000 times. In each set of 1000, do 500 with all the data, then 500 where we throw out 20% of the data.

Want a table of how many times each structure was chosen. Will have 2 tables, one for all the sets where we use all the data, then one where we only use 80%.

Table 1: 4x4 Table with Row, Column, and Overall Headers

Structure Used for Generation	**Structure Chosen**			
	Ind	**CS**	**AR1**	**Uns**
Ind	900	50	50	0
CS	Data 21	Data 22	Data 23	Data 24
AR1	Data 31	Data 32	Data 33	Data 34
Uns	Data 41	Data 42	Data 43	Data 44

With H_0 true, with the best one test the α-level (when treatment)

With $H_0 false$ (treatments no equal to each other), check power

4 groups: AIC, BIC, F-test REML, Lik Ratio ML